Politecnico di Torino

Lab report for the course Qubit Electronics

Master degree in Quantum Engineering

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In order to make the measurements as described in the Introduction section, this time the value of the central barrier gate is swept in a range that goes from 0.37 V to 0.42 V, taking 6 equally spaced values within this range:

```
V_Barrier_Vec_2 = np.linspace(0.37,0.42, 6)
```

For each value of the central barrier gate, the program solves the non-linear Poisson equation and subsequently the Schrodinger equation, in order to obtain the energy levels of the system analyzed. The workflow of the program consists of the following procedure:

- Definition of the materials assigning to each region of the system their corresponding physical properties, as in the previous laboratories.
- Set up of the parameters for solving both the Poisson and the Schrodinger equations.
- Definition of the loop that sweeps the value of the central barrier gate:

```
energy_arr = np.zeros((len(V_Barrier_Vec_2),6))
energy_arr[:,0] = V_Barrier_Vec_2
p = PoissonSolver(d, solver_params=params_poisson)

for idx, V_barrier_2 in enumerate(V_Barrier_Vec_2):
    d.new_gate_bnd('surface_Plunger_R_X_gate', V_Plunger_Right_Vec[idx] , mWF)
    d.new_gate_bnd('surface_Barrier_M_X_gate', V_barrier_2 , mWF)


    p.solve()
    d.set_V_from_phi()
    submesh = SubMesh(d.mesh, dot_region)
    dot = SubDevice(d,submesh)
    s = SchrodingerSolver(dot, solver_params=params_schrod)
    s.solve()


    energy_arr[idx,1:] = dot.energies/ct.e
    dot.print_energies()
```

What we obtain at the end, in the matrix *energy_arr*, are the first five energy eigenvalues given by the Schrödinger equation for each physical setup corresponding to the different values of the central barrier gate. After that, the program calculates the difference between the energy of the ground state and the energy of the first excited state ($E_1 - E_0$).

```
splitting = energy_arr[:,2] - energy_arr[:,1]
```

This splitting represents the energy distance between the ground states of the two separate quantum dots for each value of the barrier.

3. Analysis of the results

3.1 Theoretical framework

To properly interpret the numerical results of the simulated double quantum dot, it is essential to establish the underlying physical framework of the system. We describe the system using a localized basis consisting of the ground state wavefunctions of the individual, isolated quantum dots. Let $|L\rangle$ and $|R\rangle$ represent the states strictly localized in the left and right dot, with corresponding spatial wavefunctions $\psi_L(\mathbf{r})$ and $\psi_R(\mathbf{r})$. By projecting the full Hamiltonian $\hat{H}$ on this two dimensional subspace, we obtain the following matrix representation:

$$H=\begin{pmatrix}\langle L|\hat{H}|L\rangle & \langle L|\hat{H}|R\rangle \\ \langle R|\hat{H}|L\rangle & \langle R|\hat{H}|R\rangle\end{pmatrix}=\begin{pmatrix}\varepsilon_L & t \\ t^* & \varepsilon_R\end{pmatrix} \tag{3.1}$$

where ε_L and ε_R are the energies (eigenvalues) of the isolated left and right states, so the energies of the two dots. The off-diagonal term t. called tunneling matrix element, represents the coupling between the two states, that it how much the two eigenfunctions overlap. This latter term is crucial for understanding the behavior of the system in the analyzed regime, where the barrier between the two dots is reduced, allowing for a coupling between the two states.

The t element is defined as:

$$t=\langle L|\hat{H}|R\rangle\approx\int\psi_L^*(\mathbf{r})\Delta V(\mathbf{r})\psi_R(\mathbf{r})d\mathbf{r} \tag{3.2}$$

where $\Delta V(\mathbf{r})$ is the potential barrier separating the two dots. This integral is strictly non-zero only because the exponentially decaying tails of ψ_L and ψ_R overlap within the classically forbidden barrier region, so the region where the barrier between them is the highest.

To obtain a quantitative, analytical understanding of t, we rely on the Wentzel-Kramers-Brillouin (WKB) approximation. According to WKB theory, the transmission probability through a potential barrier scales exponentially with the barrier's geometric and energetic profile. For an energy barrier of effective width w and height V_b, the tunnel coupling can be approximated as:

$$t\propto\exp\left(-\frac{w}{\hbar}\sqrt{2m^*V_b}\right) \tag{3.3}$$

In our specific situation, the barrier height V_b is linearly modulated by the voltage applied to the central barrier gate, V_G. Using the lever-arm parameter α, the effective barrier height is $V_b(V_G)=V_{b,0}-e\alpha V_G$. Substituting this into the WKB expression reveals that the tunnel coupling grows exponentially as the applied gate voltage increases (i.e., as the barrier is lowered). When the system is tuned to perfect resonance ($\varepsilon_L=\varepsilon_R=\varepsilon_0$) using the plunger gates, the Hamiltonian becomes symmetric. Diagonalizing this matrix yields two hybridized eigenstates: a symmetric bonding state, with an eigenvalue of $E_0=\varepsilon_0-|t|$ and an antisymmetric antibonding state with corresponding eigenvalue $E_1=\varepsilon_0+|t|$. The energy splitting between these two different eigenstates is given by:

$$\Delta E=E_1-E_0=2|t| \tag{3.4}$$

in the case of a perfectly symmetric states, but in our case where the two separated dots are not perfectly identical, the splitting can be write also as:

$$\Delta E=\sqrt{(E_b-E_a)^2+(2|t|)^2} \tag{3.5}$$

where E_a and E_b are the energies of the two isolated dots. Combining this result with the WKB approximation, we obtain the fundamental relation governing the tuning of the device:

$$\Delta E \propto \exp(\gamma \cdot V_G) \tag{3.6}$$

where γ is a positive constant grouping the structural parameters of the barrier. This theoretical model predicts that the energy splitting will exhibit a pure exponential dependence on the inter-dot barrier voltage.

3.2 Energy splittings and fitting of the results

In our simulation, the energy splittings between the ground states of the two dots are reported in the following table:

Barrier Voltage (V)	**Energy splitting (μeV)**
3.69	2.08
3.80	2.91
3.90	3.83
3.99	4.84
4.09	5.94
4.19	7.20

According to the WKB (Wentzel-Kramers-Brillouin) approximation as previously described, this energy splitting follows an exponential growth as the barrier height decreases. The python script used in this laboratory and commented here aims to verify this behavior by fitting the six numerical data points with an exponential function, defined as follows:

```python
def exp_fun(x, a, b):
    return a * np.exp(b * x)

param_opt, param_cov = curve_fit(exp_fun, x_data, y_data, p0=[y_data[0], 10])
```

Listing 3.1: Exponential fit function and curve fitting procedure.

The showed function has been optimized to best fit the experimental data and the parameters a and b have been determined, resulting in $5.11e{-}10$ and 22.79 respectively.
Finally, the optimized fitting curve was plotted alongside the six numerical data points to evaluate the consistency and accuracy of the WKB approximation for this specific system configuration.

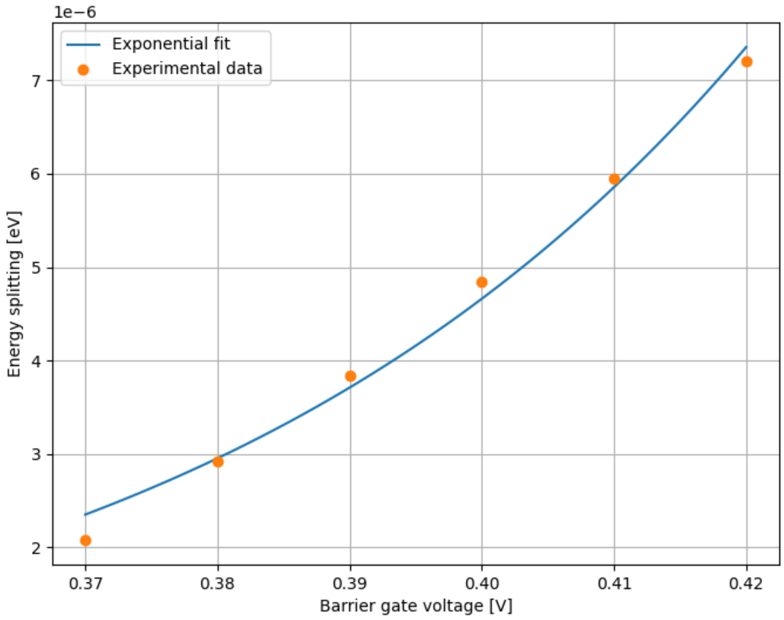

Figure 3.1: Plot of the exponential fit alongside the six numerical data points, illustrating the increase in energy splitting as the central barrier height decreases.

As shown in Fig. 3.1, the exponential fit matches the numerical data reasonably well. However, despite the data points appearing somewhat linear, this is primarily due to the limited number of sampled points, which is insufficient to fully resolve the true exponential behavior over a wider range.